

## FITZROY CROSSING, Australia

WMO: 942060

Lat: 18.1814S

Lon: 125.5619E

Elev: 115

StdP: 99.95

Time Zone: 8.00 (E08)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 9.1        | 10.7 | -10.4                         | 1.6 | 28.1 | -8.1 | 1.9 | 27.4 | 8.1                   | 23.9 | 7.4 | 24.6 | 1.3                   | 110  | 0.445 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 11            | 14.2                   | 42.5            | 20.6 | 41.4 | 20.9 | 40.4 | 21.2 | 27.1                | 32.8 | 26.5 | 32.3 | 26.1 | 31.9 | 3.9                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.9                            | 21.5 | 28.8 | 25.2 | 20.7 | 28.4 | 24.7 | 20.0 | 28.2 | 86.3          | 32.8 | 83.6 | 32.2 | 81.6 | 31.9 | 31.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |      |     |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|------|-----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |      |     |
| (4) 8.3           | 7.3  | 6.5 | DB                         | 6.1 | 44.4               | 1.5 | 0.9                                                | 5.0 | 45.0       | 4.1 | 45.6       | 3.3 | 46.1       | 2.2  | 46.7 | (4) |
| (5)               |      |     | WB                         | 2.3 | 28.0               | 1.2 | 1.4                                                | 1.4 | 29.0       | 0.7 | 29.8       | 0.1 | 30.6       | -0.8 | 31.6 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 28.0  | 31.2  | 30.6  | 30.3  | 28.7  | 24.7  | 21.7  | 21.6  | 23.2  | 27.8  | 31.6  | 32.9  | 32.2 | (6)  |
| (7)  |                                                                                  | DBStd       | 4.68  | 2.23  | 2.05  | 1.85  | 2.30  | 2.89  | 2.91  | 2.68  | 2.66  | 2.58  | 2.14  | 1.91  | 2.40 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 11    | 0     | 0     | 0     | 0     | 1     | 6     | 4     | 1     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6579  | 656   | 578   | 628   | 562   | 455   | 351   | 359   | 409   | 536   | 668   | 688   | 689  | (10) |
| (11) |                                                                                  | CDD18.3     | 3549  | 397   | 344   | 370   | 312   | 198   | 107   | 106   | 152   | 286   | 410   | 438   | 431  | (11) |
| (12) |                                                                                  | CDH23.3     | 50031 | 5432  | 4513  | 4845  | 4180  | 2442  | 1329  | 1492  | 2233  | 4120  | 6244  | 6874  | 6326 | (12) |
| (13) | CDH26.7                                                                          | 29882       | 3168  | 2544  | 2753  | 2510  | 1281  | 546   | 648   | 1160  | 2541  | 4117  | 4608  | 4006  | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 3.2   | 3.5   | 3.2   | 2.9   | 2.7   | 3.0   | 3.1   | 2.9   | 3.1   | 3.3   | 3.4   | 3.7   | 3.7  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 556   | 161   | 148   | 95    | 23    | 12    | 7     | 3     | 1     | 1     | 7     | 28    | 80   | (15) |
| (16) |                                                                                  | PrecMax     | 856   | 461   | 406   | 388   | 175   | 78    | 81    | 59    | 19    | 6     | 73    | 257   | 252  | (16) |
| (17) |                                                                                  | PrecMin     | 189   | 9     | 14    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 197   | 117   | 106   | 97    | 37    | 20    | 18    | 10    | 4     | 1     | 12    | 44    | 54   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 42.4  | 41.6  | 40.8  | 39.9  | 37.3  | 34.3  | 34.2  | 36.8  | 40.1  | 42.7  | 43.6  | 44.1 | (19) |
| (20) |                                                                                  |             | MCWB  | 21.5  | 22.4  | 21.8  | 20.1  | 19.7  | 19.3  | 17.6  | 18.4  | 18.7  | 19.3  | 19.6  | 21.1 | (20) |
| (21) |                                                                                  | 2%          | DB    | 40.5  | 40.1  | 39.4  | 38.7  | 35.9  | 32.8  | 33.0  | 35.3  | 38.8  | 41.6  | 42.4  | 42.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 22.9  | 23.8  | 22.3  | 20.3  | 18.3  | 18.3  | 17.0  | 17.4  | 18.6  | 19.5  | 20.1  | 22.0 | (22) |
| (23) |                                                                                  | 5%          | DB    | 39.0  | 38.6  | 38.1  | 37.6  | 34.6  | 31.5  | 31.9  | 33.9  | 37.8  | 40.6  | 41.3  | 40.6 | (23) |
| (24) |                                                                                  |             | MCWB  | 23.8  | 24.3  | 22.7  | 20.5  | 18.3  | 17.5  | 16.6  | 16.8  | 18.1  | 19.5  | 20.7  | 22.2 | (24) |
| (25) |                                                                                  | 10%         | DB    | 37.4  | 36.6  | 36.7  | 36.4  | 33.2  | 30.0  | 30.6  | 32.6  | 36.7  | 39.5  | 40.1  | 38.9 | (25) |
| (26) |                                                                                  |             | MCWB  | 24.4  | 24.6  | 23.4  | 20.5  | 18.1  | 16.6  | 16.2  | 16.1  | 17.8  | 19.6  | 21.2  | 22.8 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 27.3  | 28.5  | 28.1  | 26.2  | 24.6  | 22.3  | 21.6  | 21.5  | 22.4  | 24.7  | 26.2  | 26.5 | (27) |
| (28) |                                                                                  |             | MCDWB | 33.6  | 34.0  | 33.0  | 31.4  | 28.3  | 28.9  | 29.2  | 31.1  | 33.5  | 32.1  | 34.2  | 32.8 | (28) |
| (29) |                                                                                  | 2%          | WB    | 26.6  | 27.0  | 27.1  | 25.0  | 22.9  | 20.7  | 19.8  | 19.7  | 21.1  | 23.6  | 25.0  | 25.9 | (29) |
| (30) |                                                                                  |             | MCDWB | 32.5  | 32.7  | 32.4  | 31.0  | 28.4  | 27.9  | 29.2  | 30.1  | 33.3  | 31.6  | 32.0  | 32.5 | (30) |
| (31) |                                                                                  | 5%          | WB    | 26.2  | 26.4  | 26.3  | 24.2  | 21.4  | 19.5  | 18.4  | 18.5  | 20.3  | 23.0  | 24.3  | 25.4 | (31) |
| (32) |                                                                                  |             | MCDWB | 32.0  | 32.1  | 31.9  | 30.9  | 29.0  | 27.9  | 28.4  | 30.5  | 32.8  | 32.2  | 32.2  | 32.0 | (32) |
| (33) |                                                                                  | 10%         | WB    | 25.8  | 26.0  | 25.7  | 23.5  | 20.3  | 18.2  | 17.2  | 17.3  | 19.3  | 22.3  | 23.7  | 25.0 | (33) |
| (34) |                                                                                  |             | MCDWB | 31.6  | 31.7  | 31.3  | 30.6  | 29.3  | 27.9  | 28.1  | 29.9  | 32.3  | 32.4  | 32.3  | 31.6 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 11.3  | 11.3  | 12.1  | 15.3  | 16.1  | 16.5  | 17.7  | 18.8  | 17.7  | 15.8  | 14.2  | 12.5 | (35) |
| (36) |                                                                                  |             | MCDBR | 14.1  | 14.4  | 14.9  | 16.3  | 17.5  | 17.0  | 18.7  | 19.4  | 18.1  | 17.0  | 15.7  | 14.9 | (36) |
| (37) |                                                                                  |             | MCWBR | 3.8   | 4.3   | 4.2   | 5.0   | 6.1   | 6.8   | 7.4   | 7.4   | 6.2   | 5.4   | 4.6   | 3.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 10.9  | 11.2  | 11.4  | 12.3  | 12.9  | 13.8  | 15.4  | 16.5  | 15.8  | 14.2  | 13.5  | 11.8 | (38) |
| (39) |                                                                                  |             | MCWBR | 3.4   | 3.9   | 4.3   | 4.2   | 4.8   | 5.9   | 6.5   | 6.5   | 6.0   | 4.6   | 4.3   | 3.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.406 | 0.400 | 0.390 | 0.372 | 0.360 | 0.340 | 0.344 | 0.357 | 0.409 | 0.418 | 0.422 | 0.411 | (40) |      |
| (41) |                                                                                  | taud        | 2.459 | 2.480 | 2.495 | 2.474 | 2.441 | 2.471 | 2.435 | 2.360 | 2.262 | 2.280 | 2.327 | 2.422 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 939   | 935   | 921   | 901   | 873   | 876   | 881   | 901   | 889   | 907   | 918   | 934   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 121   | 117   | 111   | 106   | 102   | 95    | 101   | 117   | 138   | 141   | 137   | 125   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.74  | 6.70  | 6.25  | 5.98  | 5.16  | 4.85  | 5.14  | 6.05  | 6.93  | 7.42  | 7.47  | 7.02  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.46  | 0.57  | 0.65  | 0.24  | 0.35  | 0.26  | 0.30  | 0.18  | 0.19  | 0.30  | 0.30  | 0.42  | (45) |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only          | N/A   | +1.43   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page